R1

```
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        10.0.0.0/8 is directly connected, Ethernet0/0
L        10.0.0.1/32 is directly connected, Ethernet0/0
S     15.0.0.0/8 [1/0] via 20.0.0.2
      20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        20.0.0.0/8 is directly connected, Ethernet0/1
L        20.0.0.1/32 is directly connected, Ethernet0/1
S     21.0.0.0/8 [1/0] via 20.0.0.2
S     30.0.0.0/8 [1/0] via 20.0.0.2
S     162.0.0.0/8 [1/0] via 20.0.0.2
S     172.0.0.0/8 [1/0] via 20.0.0.2
S     192.0.0.0/8 [1/0] via 20.0.0.2
S     192.0.0.0/24 [1/0] via 20.0.0.2
R1#
R1#
```

```
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        10.0.0.0/8 is directly connected, Ethernet0/0
L        10.0.0.1/32 is directly connected, Ethernet0/0
S     15.0.0.0/8 [1/0] via 20.0.0.2
      20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        20.0.0.0/8 is directly connected, Ethernet0/1
L        20.0.0.1/32 is directly connected, Ethernet0/1
S     21.0.0.0/8 [1/0] via 20.0.0.2
S     30.0.0.0/8 [1/0] via 20.0.0.2
S     162.0.0.0/8 [1/0] via 20.0.0.2
S     172.0.0.0/8 [1/0] via 20.0.0.2
S     192.0.0.0/8 [1/0] via 20.0.0.2
S     192.0.0.0/24 [1/0] via 20.0.0.2
R1#
```

R3

```
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S     10.0.0.0/8 [1/0] via 30.0.0.1
S     15.0.0.0/8 [1/0] via 172.19.0.2
S     20.0.0.0/8 [1/0] via 30.0.0.1
      21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        21.0.0.0/8 is directly connected, Ethernet0/2
L        21.0.0.1/32 is directly connected, Ethernet0/2
      30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        30.0.0.0/8 is directly connected, Ethernet0/0
L        30.0.0.2/32 is directly connected, Ethernet0/0
S     162.0.0.0/8 [1/0] via 21.0.0.2
      172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C        172.19.0.0/16 is directly connected, Ethernet0/1
L        172.19.0.1/32 is directly connected, Ethernet0/1
S     192.0.0.0/8 [1/0] via 21.0.0.2
R3#
R3#
```

R4

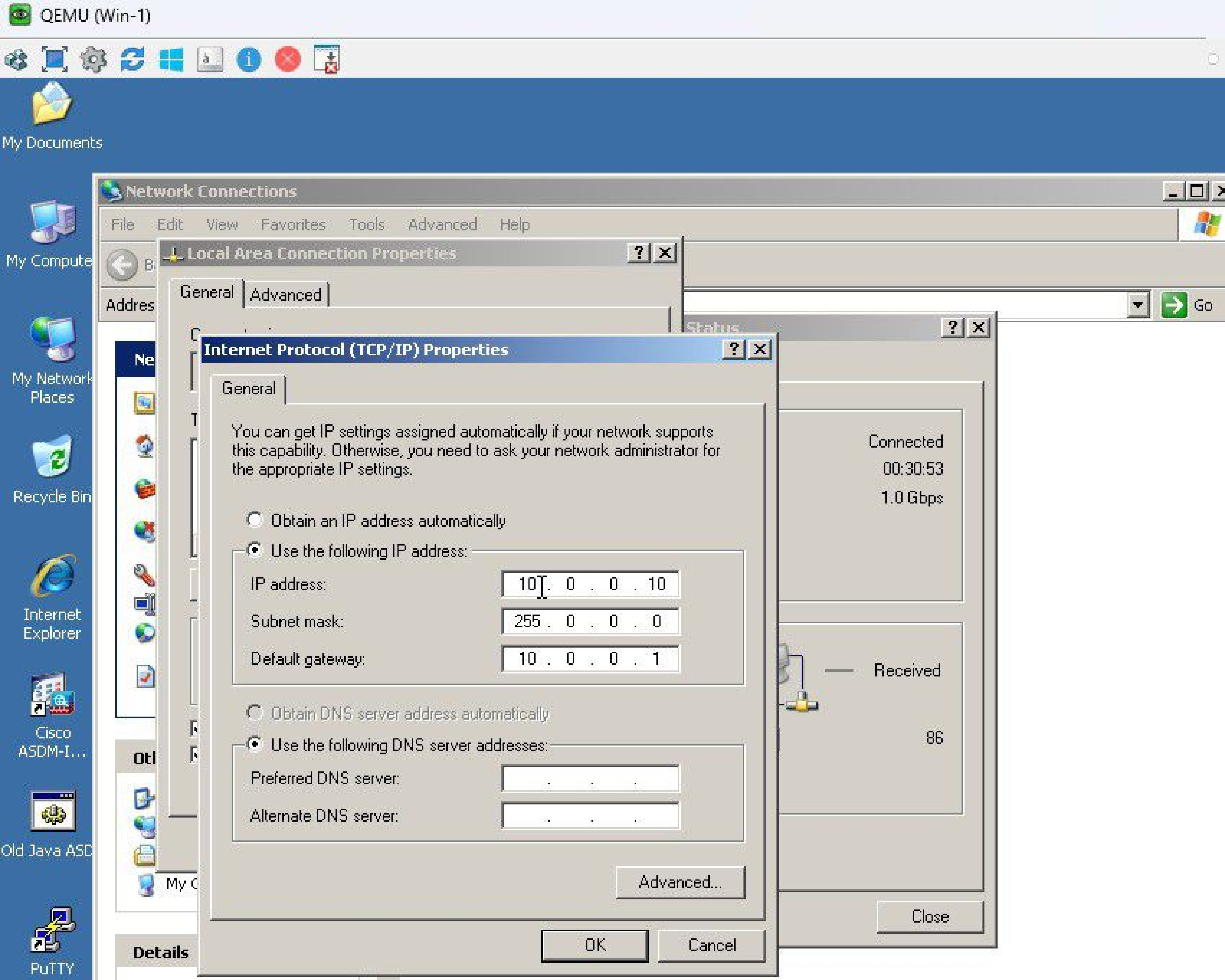
```
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S     10.0.0.0/8 [1/0] via 172.19.0.1
      15.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        15.0.0.0/8 is directly connected, Ethernet0/0
L        15.0.0.1/32 is directly connected, Ethernet0/0
S     20.0.0.0/8 [1/0] via 172.19.0.1
S     21.0.0.0/8 [1/0] via 172.19.0.1
S     30.0.0.0/8 [1/0] via 172.19.0.1
S     162.0.0.0/16 [1/0] via 172.19.0.1
      172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C        172.19.0.0/16 is directly connected, Ethernet0/1
L        172.19.0.2/32 is directly connected, Ethernet0/1
S     192.0.0.0/24 [1/0] via 172.19.0.1
R4#
R4#
```

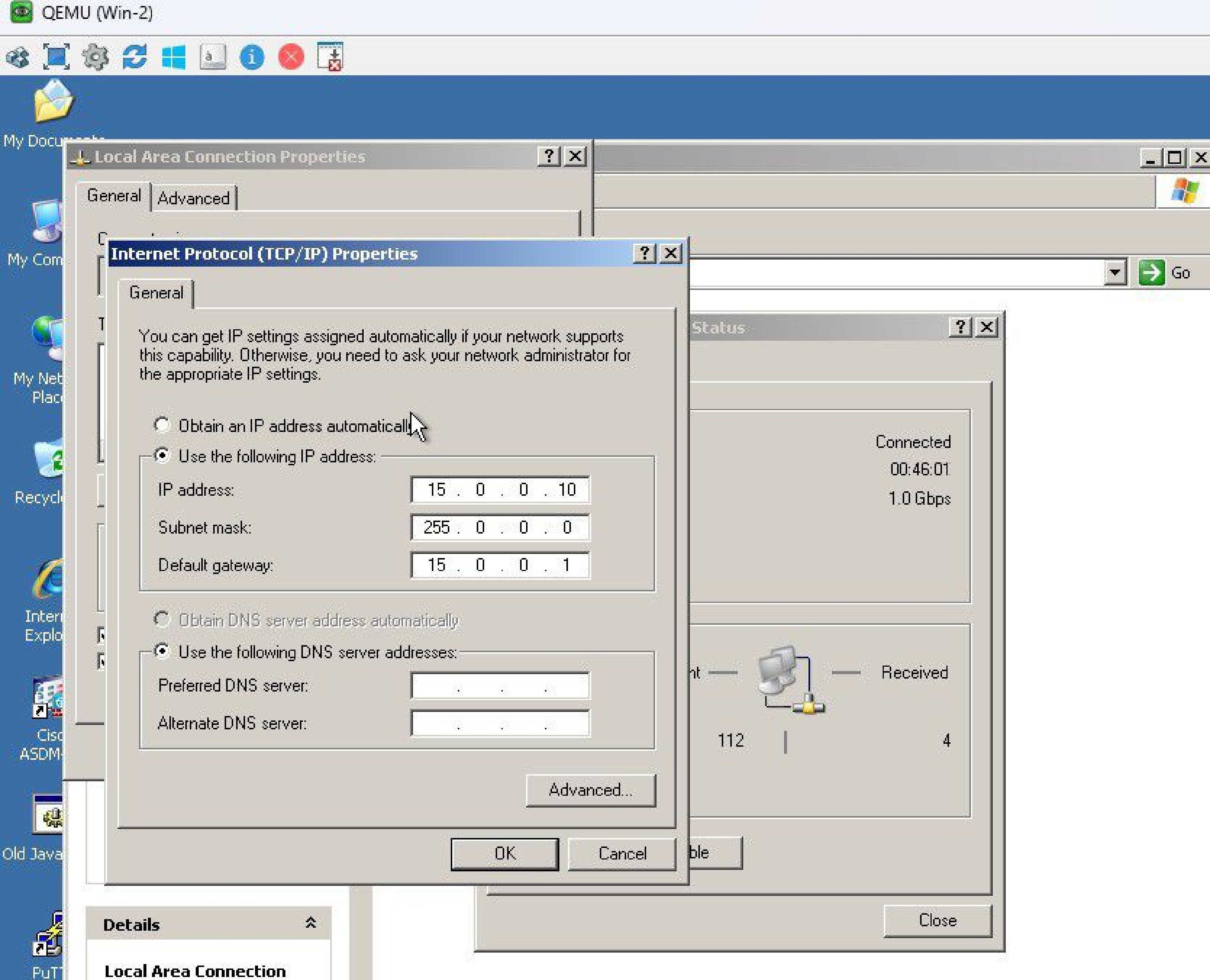
R5

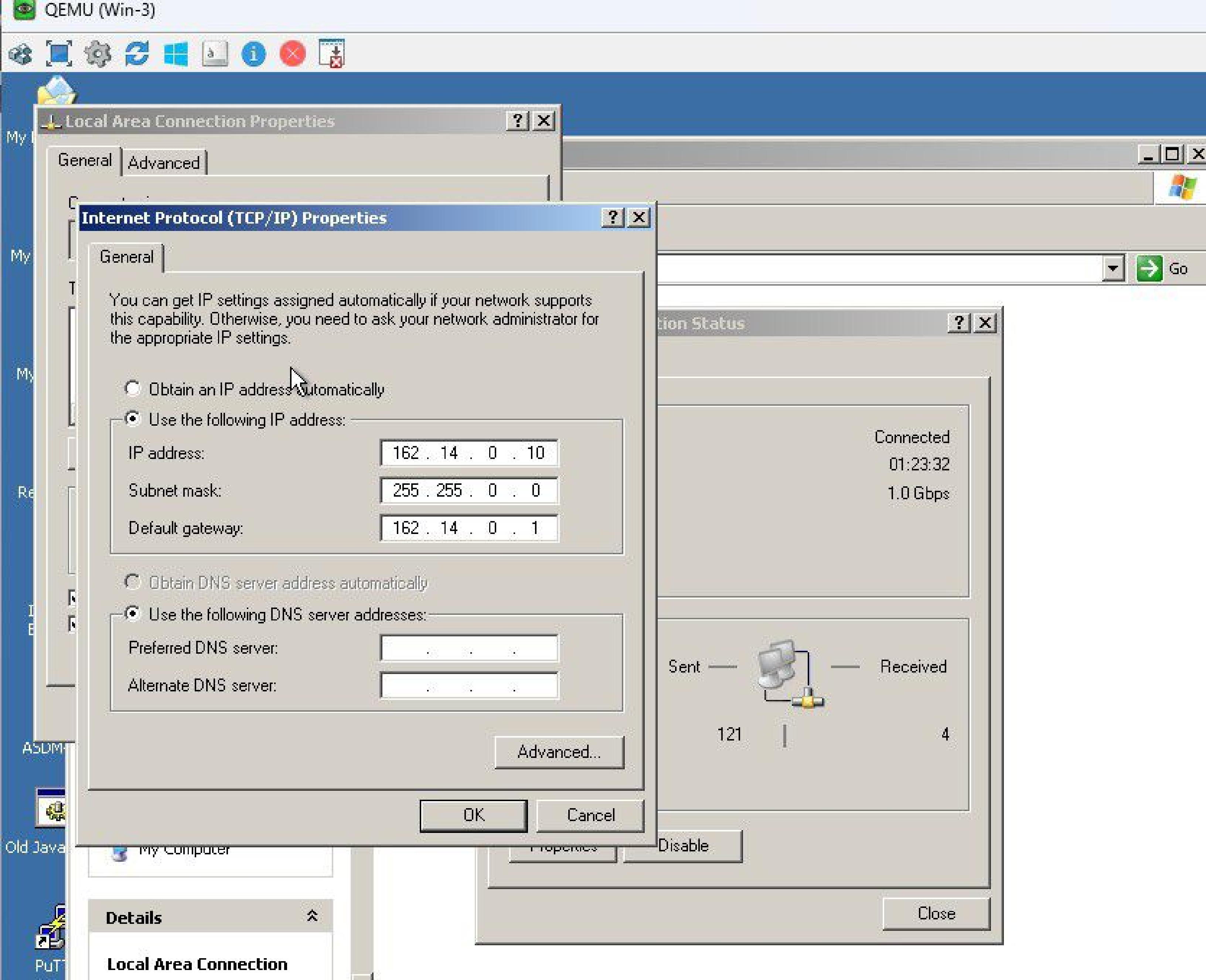
```
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S     10.0.0.0/8 [1/0] via 21.0.0.1
S     15.0.0.0/8 [1/0] via 21.0.0.1
S     20.0.0.0/8 [1/0] via 21.0.0.1
      21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        21.0.0.0/8 is directly connected, Ethernet0/0
L        21.0.0.2/32 is directly connected, Ethernet0/0
S     30.0.0.0/8 [1/0] via 21.0.0.1
S     162.0.0.0/8 [1/0] via 192.168.168.2
S     172.0.0.0/16 [1/0] via 21.0.0.1
      192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.168.0/24 is directly connected, Ethernet0/1
L        192.168.168.1/32 is directly connected, Ethernet0/1
R5#
```

R6

```
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S     10.0.0.0/8 [1/0] via 192.168.168.1
S     15.0.0.0/8 [1/0] via 192.168.168.1
S     20.0.0.0/8 [1/0] via 192.168.168.1
S     21.0.0.0/8 [1/0] via 192.168.168.1
S     30.0.0.0/8 [1/0] via 192.168.168.1
      162.14.0.0/16 is variably subnetted, 2 subnets, 2 masks
C        162.14.0.0/16 is directly connected, Ethernet0/0
L        162.14.0.1/32 is directly connected, Ethernet0/0
S     172.0.0.0/8 [1/0] via 192.168.168.1
      192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.168.0/24 is directly connected, Ethernet0/1
L        192.168.168.2/32 is directly connected, Ethernet0/1
R6#
R6#
R6#
```


QEMU (Win-1)
My Documents
My Compute
My Network Places
Recycle Bin
Internet Explorer
Cisco ASDM-I...
Old Java ASD
PuTTY
Network Connections
File Edit View Favorites Tools Advanced Help
Local Area Connection Properties
General Advanced
Go
Status
Internet Protocol (TCP/IP) Properties
General
You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.
Obtain an IP address automatically
Use the following IP address:
IP address: 10 . 0 . 0 . 10
Subnet mask: 255 . 0 . 0 . 0
Default gateway: 10 . 0 . 0 . 1
Obtain DNS server address automatically
Use the following DNS server addresses:
Preferred DNS server:
Alternate DNS server:
Advanced...
OK
Cancel
Connected
00:30:53
1.0 Gbps
Received
86
Close
Details


QEMU (Win-2)
My Documents
My Com
My Net Plac
Recycl
Inter Explo
Cisc ASDM
Old Java
PuT
Local Area Connection Properties
General
Advanced
Internet Protocol (TCP/IP) Properties
General
You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.
Obtain an IP address automatically
Use the following IP address:
IP address: 15 . 0 . 0 . 10
Subnet mask: 255 . 0 . 0 . 0
Default gateway: 15 . 0 . 0 . 1
Obtain DNS server address automatically
Use the following DNS server addresses:
Preferred DNS server:
Alternate DNS server:
Advanced...
OK
Cancel
Go
Status
Connected
00:46:01
1.0 Gbps
Received
112
4
Close
Details
Local Area Connection


QEMU (Win-3)
Local Area Connection Properties
General
Advanced
Internet Protocol (TCP/IP) Properties
General
You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.
Obtain an IP address automatically
Use the following IP address:
IP address: 162 . 14 . 0 . 10
Subnet mask: 255 . 255 . 0 . 0
Default gateway: 162 . 14 . 0 . 1
Obtain DNS server address automatically
Use the following DNS server addresses:
Preferred DNS server:
Alternate DNS server:
Advanced...
OK
Cancel
Go
Connected
01:23:32
1.0 Gbps
Sent
Received
121
4
Disable
Close
Details
Local Area Connection

QEMU (Win-1)

C:\WINDOWS\system32\cmd.exe

```
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

Reply from 10.0.0.10: bytes=32 time<1ms TTL=128
Reply from 10.0.0.10: bytes=32 time<1ms TTL=128
Reply from 10.0.0.10: bytes=32 time<1ms TTL=128
Reply from 10.0.0.10: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Documents and Settings\user>ping 15.0.0.10

Pinging 15.0.0.10 with 32 bytes of data:

Reply from 15.0.0.10: bytes=32 time=3ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 15.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>ping 162.14.0.10

Pinging 162.14.0.10 with 32 bytes of data:

Reply from 162.14.0.10: bytes=32 time=3ms TTL=123
Reply from 162.14.0.10: bytes=32 time=2ms TTL=123
Reply from 162.14.0.10: bytes=32 time=2ms TTL=123
Reply from 162.14.0.10: bytes=32 time=2ms TTL=123

Ping statistics for 162.14.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>_
```

QEMU (Win-2)

C:\WINDOWS\system32\cmd.exe

```
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ping 162.14.0.10

Pinging 162.14.0.10 with 32 bytes of data:

Reply from 162.14.0.10: bytes=32 time=3ms TTL=124
Reply from 162.14.0.10: bytes=32 time=2ms TTL=124
Reply from 162.14.0.10: bytes=32 time=2ms TTL=124
Reply from 162.14.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 162.14.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>ping 15.0.0.10

Pinging 15.0.0.10 with 32 bytes of data:

Reply from 15.0.0.10: bytes=32 time<1ms TTL=128
Reply from 15.0.0.10: bytes=32 time<1ms TTL=128
Reply from 15.0.0.10: bytes=32 time<1ms TTL=128
Reply from 15.0.0.10: bytes=32 time<1ms TTL=128

Ping statistics for 15.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Documents and Settings\user>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

Reply from 10.0.0.10: bytes=32 time=3ms TTL=124
Reply from 10.0.0.10: bytes=32 time=3ms TTL=124
Reply from 10.0.0.10: bytes=32 time=4ms TTL=124
Reply from 10.0.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 4ms, Average = 3ms

C:\Documents and Settings\user>
```

QEMU (Win-3)

C:\WINDOWS\system32\cmd.exe

```
Reply from 162.14.0.10: bytes=32 time=1ms TTL=128
Reply from 162.14.0.10: bytes=32 time<1ms TTL=128
Reply from 162.14.0.10: bytes=32 time<1ms TTL=128
Reply from 162.14.0.10: bytes=32 time<1ms TTL=128

Ping statistics for 162.14.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Documents and Settings\user>ping 15.0.0.10

Pinging 15.0.0.10 with 32 bytes of data:

Reply from 15.0.0.10: bytes=32 time=3ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 15.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

Reply from 10.0.0.10: bytes=32 time=3ms TTL=123
Reply from 10.0.0.10: bytes=32 time=2ms TTL=123
Reply from 10.0.0.10: bytes=32 time=2ms TTL=123
Reply from 10.0.0.10: bytes=32 time=3ms TTL=123

Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>_
```